Design of an 8-Bit, 250 MS/s Binary-Weighted Current-Steering DAC in 0.18 µm CMOS

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***Abstract*—This paper presents the design of an 8-bit, 250 MS/s digital-to-analog converter (DAC) implemented in TSMC 0.18 µm CMOS technology with a 1.8 V supply. A binary-weighted current-steering architecture is adopted, consisting of eight binary-weighted NMOS bit cells with weights 1 through 128, a NAND-latch-based input retimer, a 100 µA NMOS-mirror bias generator, and a per-bit 6-bit analog weight trim DAC for mismatch correction. The differential output achieves a 1.50 V peak-to-peak swing across a 100 Ω differential / 25 Ω common-mode load. Across TT, SS, and FF process corners, the design achieves a minimum SFDR of 48.2 dB across the tested Nyquist-band input frequencies, meeting the 48 dB SFDR requirement. The design also achieves DNL < 0.18 LSB and INL < 0.24 LSB, well within the ±1/±2 LSB limits. The 6-bit trim DAC provides ±14% current tuning range per bit, exceeding the ±10% specification.**

***Index Terms*—Current-steering DAC, binary-weighted, 0.18 µm CMOS, SFDR, DNL, INL, analog weight tuning, input retimer**

I. Introduction

HIGH-SPEED digital-to-analog converters are fundamental building blocks in communication systems, arbitrary waveform generators, and software-defined radio front-ends. The purpose of this project is to design an 8-bit DAC at 250 MS/s in TSMC 0.18 µm CMOS. The remainder of this paper is organized as follows. Section II describes the architecture, Section III elaborates its building blocks, and Section IV presents the simulation results. The design specifications are summarized in Table I.

TABLE I
DESIGN SPECIFICATIONS

Parameter	Requirement	Unit
Supply Voltage	1.8	V
Sample Rate	250	MS/s
Resolution	8	bits
Output Type	Differential	–
Output Swing	$\geq$1.5	$\mathrm{V_{pp,diff}}$
SFDR	>48	dB
Diff. Output Resistance	100	Ω
CM Output Resistance	25	Ω
Load Capacitance	550	fF (each output)
DNL	$< \pm 1$	LSB
INL	$< \pm 2$	LSB
Weight Tuning Range	±10	%/bit
Reference Current	100	µA

II. Architecture

A. Binary-Weighted Current-Steering Architecture

The DAC employs a direct binary-weighted current-steering architecture. Eight NMOS bit cells with binary current weights ($2^k \times I_{\text{unit}}$, for $k = 0, 1, \ldots, 7$) steer current to either the positive or negative output node based on the corresponding digital input bit. The total full-scale output current is:

$$I_{\text{FS}} = I_{\text{unit}} \sum_{k=0}^{7} 2^k = 255\, I_{\text{unit}}. \tag{1}$$

For a 1.5 $\mathrm{V_{pp,diff}}$ swing across the 100 Ω differential load (each side presents a 50 Ω path to VDD), the required unit current is:

$$I_{\text{unit}} = \frac{V_{\text{swing}}/2}{255 \times 50\,\Omega} = \frac{0.75}{12750} \approx 58.8\,\mu\text{A}. \tag{2}$$

A binary-weighted topology was chosen over a segmented architecture because it requires only 8 active cells and no thermometer decoder, minimizing digital complexity and routing congestion.

B. Analog Weight Tuning

Device mismatch causes the actual current of each bit cell to deviate from its ideal binary weight, degrading INL and DNL. Each bit cell therefore incorporates an independent 6-bit trim DAC that provides a programmable sink current at the internal drain node. By defining trim code 32 as the nominal setting, the trim DAC provides effective positive or negative correction relative to the nominal bit-cell current, with a tuning range exceeding $\pm 10\%$ of the nominal cell current. This allows per-cell calibration to recover linearity.

C. Output Network

The differential output is terminated with two 50 Ω resistors each connected from VDD to the respective output node (OUTp and OUTn), plus 550 fF capacitors from each output to ground modeling bond-pad parasitics. This configuration yields the required 100 Ω differential impedance (two 50 Ω in series for differential mode) and 25 Ω common-mode impedance (two 50 Ω in parallel). The output common-mode voltage is constant at:

$$V_{\text{CM}} = V_{\text{DD}} - \frac{I_{\text{FS}}}{2} \times 50\,\Omega \approx 1.42\,\text{V}, \tag{3}$$

independent of input code since the total current drawn from both output nodes always equals I_{FS}.

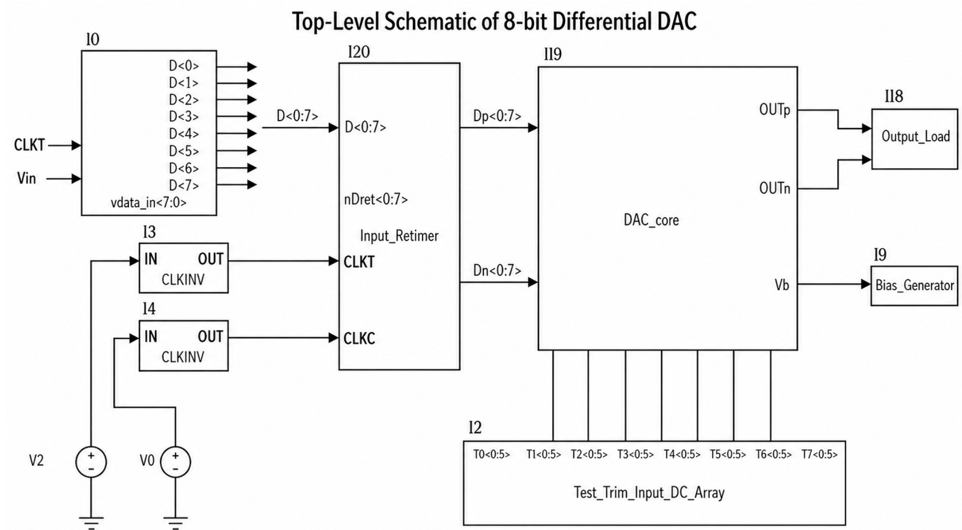

Fig. 1. Top-level schematic of the 8-bit DAC.

III. Building Blocks

A. Top-Level Design

The top-level design is shown in Fig. 1 and contains four main blocks:

- Bias Generator
- Input Retimer
- DAC Core
  - 8 Binary-Weighted Bit Cells
  - Per-bit 6-bit Trim DAC
- Output Load (50 Ω + 550 fF per node)

B. Single Bit Cell of DAC Core

The DAC core uses an 8-bit binary-weighted current-steering architecture. It contains eight bit cells with weights 1, 2, 4, 8, 16, 32, 64, and 128, corresponding to the 8-bit input code from LSB to MSB. Each bit cell steers its weighted current to either the positive or negative output branch according to the retimed differential digital inputs D_p and D_n, producing a differential analog output current. A single bit cell schematic is shown in Fig. 2.

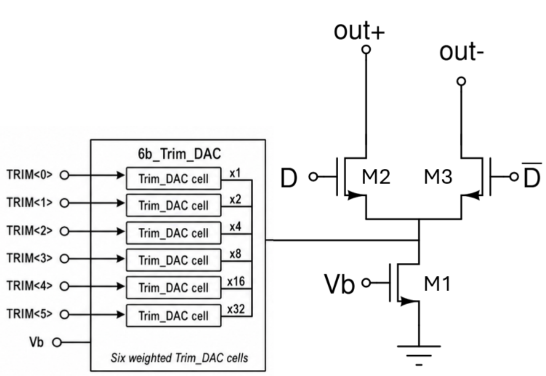

Fig. 2. Single bit DAC cell tuned by a 6-bit trim DAC.

TABLE II
Current-Steering Bit Cell Sizing

	W	L	Multiplier
M1 (Current Source)	1.183 µm	500 nm	weight
M2 (Switch P-side)	500 nm	180 nm	weight
M3 (Switch N-side)	500 nm	180 nm	weight

The multiplier is defined as $m = 2^k$ for bit k. The MSB cell ($k = 7$) uses $m = 128$ fingers and the LSB cell ($k = 0$) uses a single finger. All fingers share the same V_b gate bias, so the current scales with multiplicity in the absence of mismatch. For example, the effective switch width of the MSB cell is:

$$W_{\text{eff, MSB}} = 500\,\text{nm} \times 128 = 64\,\mu\text{m}. \tag{4}$$

The relative current mismatch of the unit cell is:

$$\frac{\sigma(\Delta I)}{I} = \sqrt{\frac{A_{VT}^2}{(V_{GS} - V_{TH})^2 \cdot WL} + \frac{A_\beta^2}{WL}}, \tag{5}$$

with $A_{VT} = 5\,\text{mV}\cdot\mu\text{m}$ and $A_\beta = 0.01\,\mu\text{m}$.

C. Trim DAC Cell

Each bit cell incorporates a 6-bit trim DAC block consisting of six binary-weighted single-bit trim DAC unit cells with weights $1, 2, 4, 8, 16, 32$ times the cell's base multiplicity $m = 2^k$.

Each single-bit trim DAC cell is realized by a long-channel NMOS current source accompanied by a transmission gate, as shown in Fig. 3.

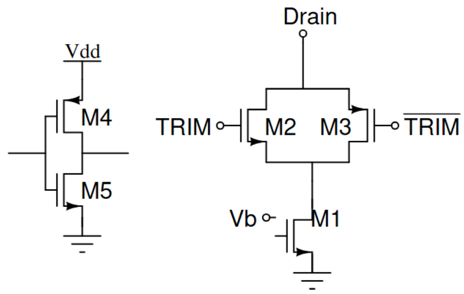

Fig. 3. Single-bit trim DAC cell.

TABLE III
TRIM DAC CELL SIZING

	W	L	Multiplier
M1 (Trim Source)	220 nm	20 µm	weight
M2 (Gate P)	4.5 µm	180 nm	2
M3 (Gate N)	4.5 µm	180 nm	1
M4	10 µm	180 nm	1
M5	10 µm	180 nm	1

The long channel ($L = 20\,\mu$m) of the trim current source provides high output impedance, ensuring precise code-independent correction currents. The TRIM signal is set to 100000_2 (decimal 32) by default, which turns on only the MSB cell. Turning off the MSB reduces the current by 10%; turning on the remaining five cells increases it by 10%, realizing a ±10% range.

The completed 8-bit DAC core is shown in Fig. 4.

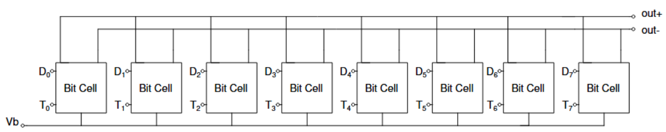

Fig. 4. Complete 8-bit DAC core.

D. Input Retimer

To ensure all bit cells switch simultaneously and suppress inter-bit skew, all 8 data bits are re-timed to the rising edge of CLK_T by a dedicated Input Retimer block. Each bit passes through a NAND-latch-based retimer cell, shown in Fig. 5. The advantage of NAND-latch logic is that it produces synchronous complementary outputs D_p and D_n, which suppresses even-order harmonics (HD2) and improves SFDR.

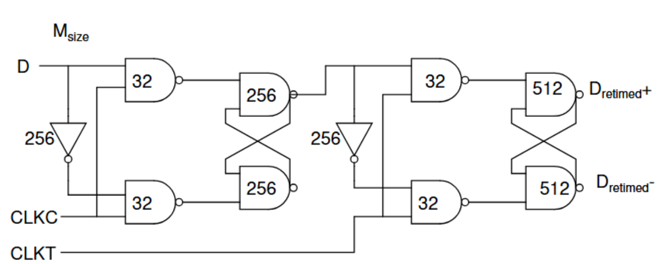

Fig. 5. Single-bit retimer cell (NAND-latch based). Msize's, the sizing parameters, are labelled on each components

The NAND gate implementation is shown in Fig. 6.

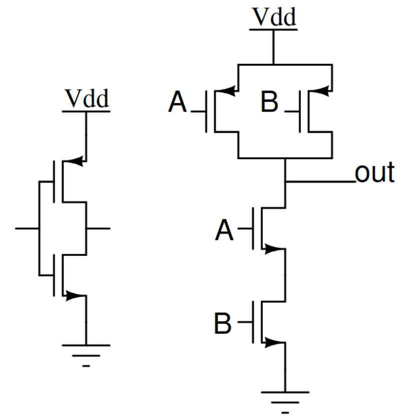

Fig. 6. NAND gate schematic.

NAND gate near output are using larger sizing to minimize delay and improve drivability. The complete retimer block,

TABLE IV
INPUT RETIMER NAND AND INVERTER SIZING

	W	L	Multiplier
NAND PMOS	220 nm	180 nm	2*Msize
NAND NMOS	220 nm	180 nm	2*Msize
INV PMOS	220 nm	180 nm	2*Msize
INV NMOS	220 nm	180 nm	Msize

shown in Fig. 7, contains 8 single-bit retimer cells, one per data bit.

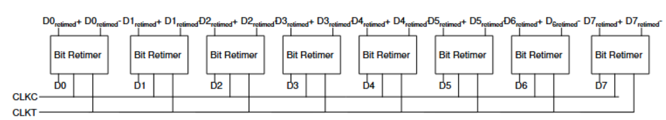

Fig. 7. Complete 8-bit input retimer block.

E. Bias Generator

All current sources share a common gate bias V_b generated by forcing 100 µA through a diode-connected NMOS transistor, shown in Fig. 8.

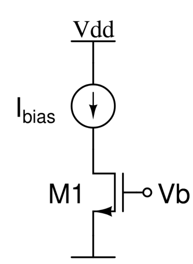

Fig. 8. Bias generator.

V_b is distributed to every bit cell, ensuring all currents track each other over process and temperature variations.

TABLE V
BIAS GENERATOR PARAMETERS

I_{bias}	100 µA
V_{DD}	1.8 V
W	3.5 um
L	800 nm
Multiplier	1

IV. Simulation Results

A. Simulation Setup

Static linearity is evaluated by sweeping all 256 digital codes in a DC parametric simulation and recording the differential output voltage V_{diff} and single-ended outputs $V_{\text{OUTp}}/V_{\text{OUTn}}$. INL and DNL are computed as:

$$\text{DNL}[k] = \frac{V_{\text{diff}}[k+1] - V_{\text{diff}}[k]}{\text{LSB}} - 1, \quad \text{INL}[k] = \frac{V_{\text{diff}}[k] - V_{\text{ideal}}[k]}{\text{LSB}}. \tag{6}$$

Dynamic performance is evaluated using six coherent input tones with M = 2048 samples and J = 17, 127, 251, 503, 751, and 997, where fin = Jfs/M. The exported output spectrum is post-processed with zero-order-hold sinc correction before calculating SFDR, SNDR, and ENOB. DC bins and the fundamental ±2 bins are excluded when identifying the largest spur, while the fundamental power is integrated over ±2 bins for SNDR and SFDR calculation. All simulations run at $T = 27°$C across TT, SS, and FF corners.

B. Output Impedance

The output impedance is characterized by an AC frequency sweep at each of the $2^8 = 256$ input codes, producing R_{out} per code. The low-frequency (DC) output resistance is extracted and shown in Fig. 9 which plots $R_{\text{out,diff}}$ and $R_{\text{out,CM}}$ at low frequency against the 8-bit binary input code.

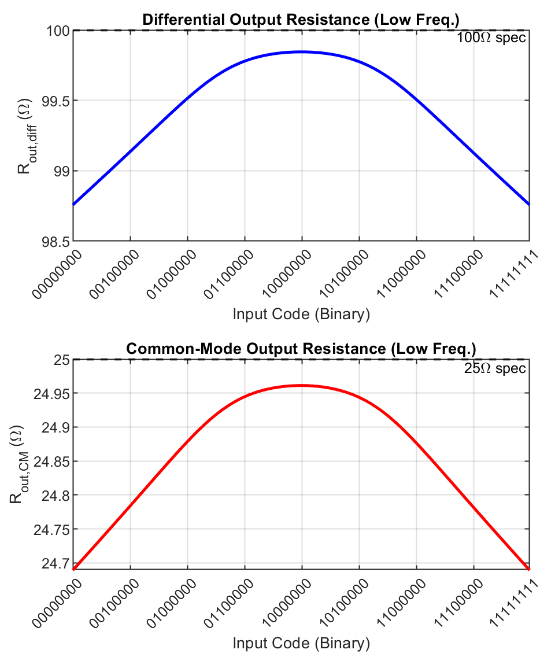

Fig. 9. Low-frequency $R_{\text{out,diff}}$ (top) and $R_{\text{out,CM}}$ (bottom) vs. 8-bit binary input code. Dashed lines mark the 100 Ω and 25 Ω specifications.

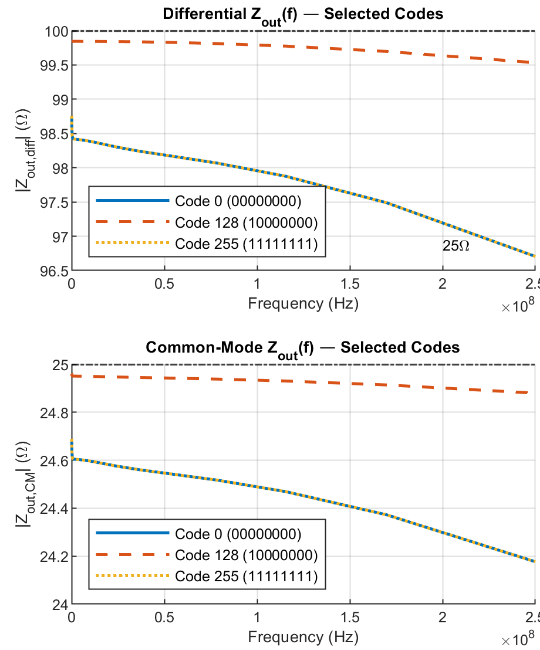

Fig. 10. $|Z_{\text{out}}(f)|$ vs. frequency for codes 0, 128, and 255 — differential (top) and common-mode (bottom).

The differential output resistance varies only from 98.76 Ω to 99.84 Ω across all 256 codes (<1.1% variation), and the common-mode resistance from 24.69 Ω to 24.96 Ω (<1.1% variation). Both are within 1.3% of their respective 100 Ω and 25 Ω targets, confirming the specifications are met across the full code range. The tiny code dependence arises from the finite output impedance r_o of the NMOS current sources appearing in parallel with the 50 Ω load resistors; since $r_o \gg 50\,\Omega$ for these devices, the effect is negligible.

C. Output Swing and Common-Mode Voltage

Table VI shows the measured differential peak-to-peak swing and common-mode voltage across process corners. All corners achieve $V_{\text{diff,pp}} \geq 1.50$ V, meeting the 1.5 V specification. The common-mode voltage is ≈1.42 V, close to the theoretical value $V_{\text{CM}} = 1.8 - (255 \times 59.2\,\mu\text{A}/2) \times 50\,\Omega = 1.42$ V.

TABLE VI
OUTPUT SWING ACROSS PROCESS CORNERS

Corner	$V_{\text{diff,pp}}$ (V)	$V_{\text{OUTp,pp}}$ (V)	$V_{\text{OUTn,pp}}$ (V)	V_{CM} (V)
SS	1.501	0.751	0.751	1.424
TT	1.502	0.751	0.751	1.423
FF	1.501	0.750	0.750	1.423

D. Retimer Timing Verification

The input retimer re-times all 8 data bits to the rising edge of CLK_T, ensuring simultaneous switching of all current cells. Fig. 11 shows the input data $D\langle 0\rangle$, the retimed output $D_{\text{ret}}\langle 0\rangle$, and both clock phases over a 40 ns window (10 clock cycles at 250 MHz). Fig. 12 zooms in to a single transition to annotate the CLK_T-to-D_{ret} propagation delay.

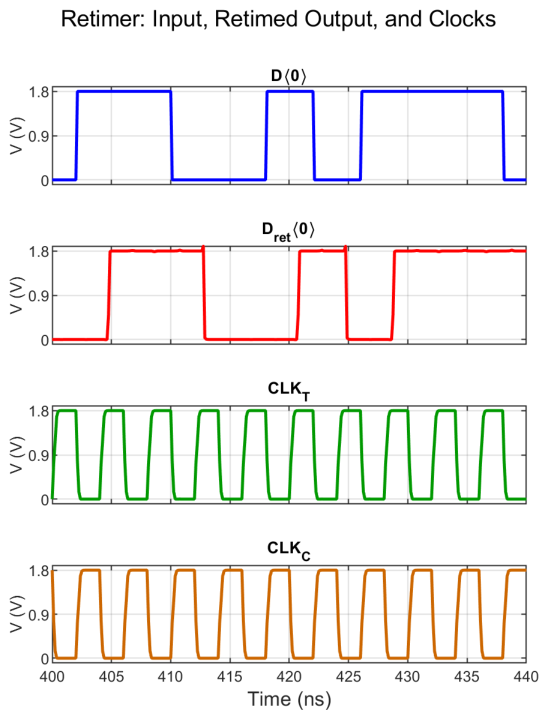

Fig. 11. Retimer waveforms: input $D\langle 0\rangle$, retimed output $D_{\text{ret}}\langle 0\rangle$, CLK_T, and CLK_C over 40 ns.

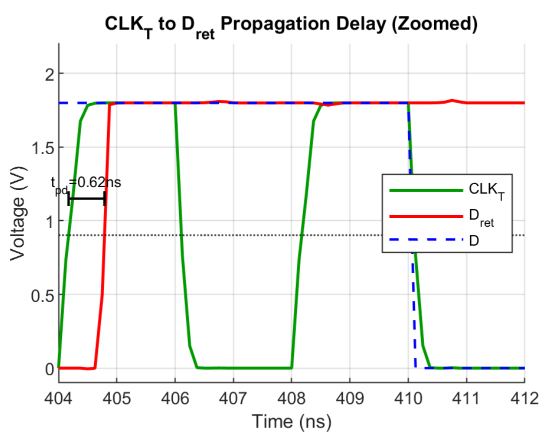

Fig. 12. Zoomed view showing the CLK_T-to-D_{ret} propagation delay annotated at the 50% crossing.

TABLE VII
RETIMER TIMING METRICS (TT CORNER, 250 MHZ)

Parameter	Value
CLK_T $\rightarrow$ D_{ret} rising-edge delay	0.616 ns
CLK_T $\rightarrow$ D_{ret} falling-edge delay	0.644 ns
Average propagation delay t_{pd}	0.627 ns
t_{pd} as fraction of clock period	15.7 %
D_{ret} rise slew (10%–90%)	7.63 V/ns
D_{ret} fall slew (90%–10%)	14.90 V/ns

The average propagation delay of 0.627 ns is 15.7% of the 4 ns clock period, leaving ample setup margin for the current-cell switches. The falling edge is faster than the rising edge (14.9 V/ns vs. 7.6 V/ns), consistent with the stronger NMOS pull-down path in the output buffer inverters. Both edges are fully settled well within one clock cycle, confirming that D_{ret} presents a clean, synchronous signal to the DAC current-cell switches.

E. Static Linearity — INL and DNL

Table. VIII shows INL and DNL across all three process corners. Peak values are summarized in Table VIII.

TABLE VIII
PEAK INL AND DNL ACROSS PROCESS CORNERS

Corner	DNL_{min} (LSB)	DNL_{max} (LSB)	INL_{min} (LSB)	INL_{max} (LSB)
SS	-0.172	$+0.002$	-0.086	$+0.086$
TT	-0.026	$+0.003$	-0.159	$+0.159$
FF	-0.012	$+0.005$	-0.235	$+0.235$
Requirement	> -1	$< +1$	> -2	$< +2$

All corners are well within the ± 1 LSB DNL and ± 2 LSB INL limits. The peak INL magnitude of 0.235 LSB (FF) is $10\times$ better than required. The systematic increase of INL from SS to FF reflects the change in device transconductance with process corner, which shifts the effective binary weights slightly; the monotonic INL shape confirms no missing codes.

F. Analog Weight Tuning Range

Per-bit current deviation as the 6-bit trim code is swept from 0 to 63 (reference = trim 32).

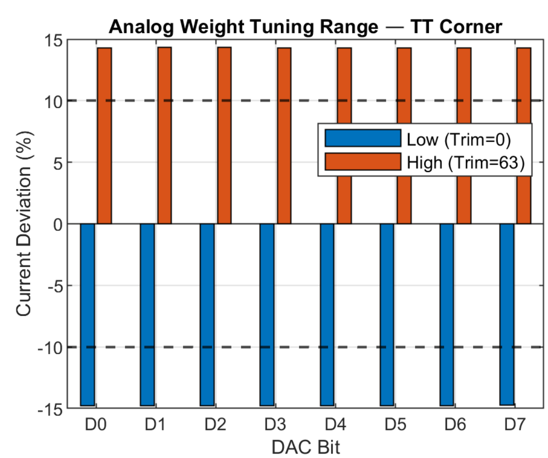

Fig. 13. Analog weight tuning range per bit in the TT corner. Dashed lines mark the $\pm 10\%$ specification.

The measured tuning range is -14.8% (trim = 0) to $+14.3\%$ (trim = 63), averaged across all 8 bits. This exceeds the $\pm 10\%$ specification with >40% margin. The tuning range is identical for bits D0–D6 and nearly identical for D7, confirming proper binary scaling of the trim cells.

G. Dynamic Performance

Output Spectra: Fig. 14 shows the output spectrum at low input frequency ($J = 17$, $f_{\text{in}} = 2.07\,\text{MHz}$, TT corner). Fig. 15 shows the spectrum near Nyquist ($J = 997$, $f_{\text{in}} = 121.7\,\text{MHz}$, TT corner).

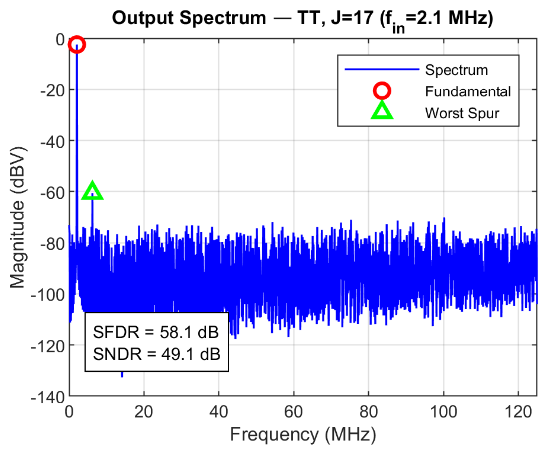

Fig. 14. Output spectrum at $f_{\text{in}} = 2.07\,\text{MHz}$ (TT). SFDR = 58.1 dB, SNDR = 49.1 dB.

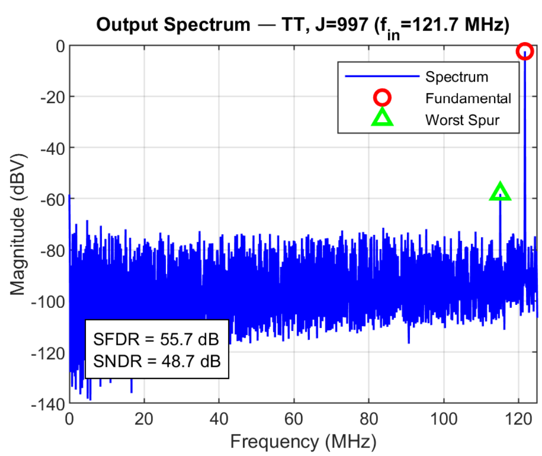

Fig. 15. Output spectrum at $f_{\text{in}} = 121.7\,\text{MHz}$ (TT). SFDR = 55.7 dB, SNDR = 48.7 dB.

SFDR and SNDR vs. Input Frequency: Fig. 16 shows SFDR as a function of input frequency across all three process corners; Fig. 17 shows the corresponding SNDR.

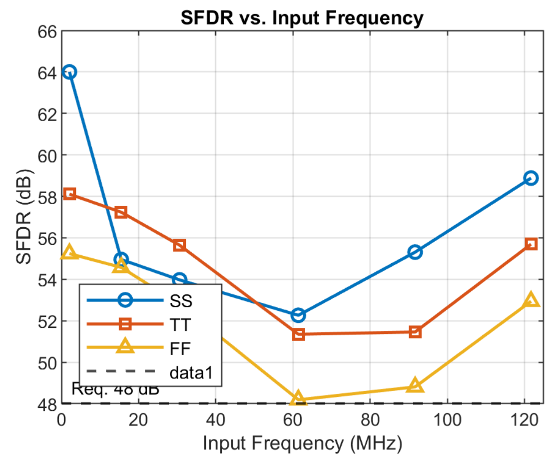

Fig. 16. SFDR vs. input frequency across process corners. The dashed line marks the 48 dB requirement.

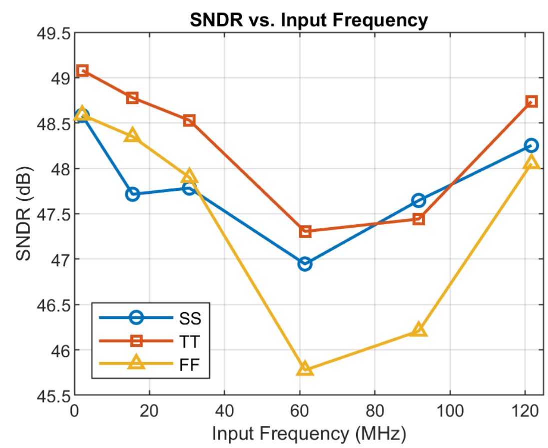

Fig. 17. SNDR vs. input frequency across process corners.

The TT corner achieves SFDR of 51–58 dB and SNDR of 47–49 dB across the Nyquist band. The FF corner exhibits the worst SFDR, dropping to 48.2 dB at $f_{\text{in}} \approx 61\,\text{MHz}$ — the minimum across all corners and frequencies, which just meets the 48 dB specification.

H. Performance Summary

Table IX summarizes the key simulated performance metrics.

I. Power Consumption

Power is measured by time-averaging the instantaneous supply power from 10 ns to 990 ns (one full μs window) in transient simulation at three representative input frequencies in the TT corner. Table. X shows the breakdown by block.

The DAC core and load consume nearly constant power (≈30 mW) across all input frequencies, as expected for an analog current-steering circuit. The bias generator draws only 0.18 mW. The retimer is the dominant frequency-dependent power block: its dynamic CMOS power scales with switching activity, rising from 20.9 mW at 2.1 MHz to 30.6 mW at 121.7 MHz. Total power ranges from 51.0 mW to 60.8 mW across the Nyquist band.

TABLE IX
PERFORMANCE SUMMARY ACROSS PROCESS CORNERS

Parameter	SS	TT	FF
$V_{\text{diff,pp}}$ (V)	1.501	1.502	1.501
V_{CM} (V)	1.424	1.423	1.423
DNL peak (LSB)	−0.172	−0.026	−0.012
INL peak (LSB)	±0.086	±0.159	±0.235
SFDR @ $f_{\text{in}} = 2.1$ MHz (dB)	64.0	58.1	55.2
SFDR @ $f_{\text{in}} = 121.7$ MHz (dB)	58.9	55.7	52.9
SFDR minimum across Nyquist (dB)	52.3	51.4	48.2
SNDR @ $f_{\text{in}} = 2.1$ MHz (dB)	48.6	49.1	48.6
SNDR @ $f_{\text{in}} = 121.7$ MHz (dB)	48.3	48.7	48.1
ENOB @ $f_{\text{in}} = 2.1$ MHz (bits)	7.78	7.86	7.78
ENOB @ $f_{\text{in}} = 121.7$ MHz (bits)	7.72	7.80	7.69
$R_{\text{out,diff}}$ range (Ω)		98.76–99.84	
$R_{\text{out,CM}}$ range (Ω)		24.69–24.96	
Trim tuning range (%)		-14.8 to $+14.3$	
Power — Retimer (mW) @ 2.1 MHz		20.9	
Power — Retimer (mW) @ 121.7 MHz		30.6	
Power — DAC + Load (mW)		≈30.0	
Power — Bias Generator (mW)		0.18	
Total Power (mW) @ 2.1 MHz		51.0	
Total Power (mW) @ 121.7 MHz		60.8	

TABLE X
AVERAGE POWER BY BLOCK — TT CORNER

Block	Low (2.1 MHz)	Mid (61.4 MHz)	High (121.7 MHz)
Retimer	20.85 mW	26.41 mW	30.59 mW
Bias Generator	0.18 mW	0.18 mW	0.18 mW
DAC + Load	29.93 mW	29.98 mW	30.02 mW
Total	50.96 mW	56.57 mW	60.79 mW

J. SFDR Limiting Factors

The primary SFDR bottleneck is the switching nonlinearity of the high-weight current cells (MSB and MSB-1). In a binary-weighted architecture, the MSB cell alone carries 128 LSBs of current and uses 128 parallel switch transistors; the large gate capacitance of these wide devices causes code-dependent charge injection and finite rise/fall time at the switch drains, producing harmonic distortion. FF has faster transition and stronger switching transients, which can worsen dynamic distortion. The minimum SFDR (48.2 dB in FF at 61 MHz) occurs at mid-frequency, where the aliases of even-order switching transients from the MSB cells fall inside the passband. The NAND-latch retimer suppresses HD2 by ensuring D_p and D_n switch symmetrically, but odd-order harmonics (HD3) remain, ultimately limiting SFDR.

V. CONCLUSION

An 8-bit, 250 MS/s binary-weighted current-steering DAC was designed in TSMC 0.18 µm CMOS. The design meets all static specifications: differential output swing of 1.50 $\text{V}_{\text{pp,diff}}$, DNL < 0.18 LSB, and INL < 0.24 LSB across TT, SS, and FF corners. The 6-bit analog weight trim DAC provides a $\pm14\%$ tuning range per bit, exceeding the $\pm10\%$ requirement. Dynamic performance satisfies the 48 dB SFDR target across the Nyquist band for all corners, with the tightest margin in the FF corner at mid-frequency (48.2 dB). The dominant SFDR limiter is the switching nonlinearity of the MSB current cell, driven by code-dependent charge injection in the wide switch transistors. The output impedance meets specification across all 256 input codes: the differential Rout is 98.76–99.84 Ω ($<1.1\%$ of the 100 Ω target) and the CM Rout is 24.69–24.96 Ω ($<1.3\%$ of the 25 Ω target), confirming the 50 Ω load-resistor network correctly sets both impedances. Total power ranges from 51.0 mW (at 2.1 MHz) to 60.8 mW (at 121.7 MHz), with the retimer as the dominant frequency-dependent consumer.